Modeling and Forecasting Monthly Atmospheric CO_2 at Mauna Loa, 1958–2026

A Seasonal ARIMA Analysis

Pascal Antwi

December 3, 2026

STA6938 Time Series · Final Project

Roadmap

1. Background, motivation and data
2. Trend, seasonality and stationarity
3. Model identification and selection
4. Estimation, tests and confidence intervals
5. Model adequacy and remedial measures
6. Forecast evaluation and forecasts to 2030

Research questions

- How fast is CO_2 rising?
- Is the rise accelerating?
- What levels should we expect by 2030?

Background and motivation

Rationale

- CO_2 is the principal driver of anthropogenic climate change.
- Climate targets depend on *how fast* CO_2 rises and *when* it crosses thresholds.
- Rigorous characterization and forecasting are of direct scientific and policy relevance.
- Methodologically rich: stochastic trend, seasonal cycle and serial dependence.

The question

CO_2 is rising; the question is **how fast**, **whether it is accelerating**, and **what to expect next**, with proper uncertainty.

Objectives

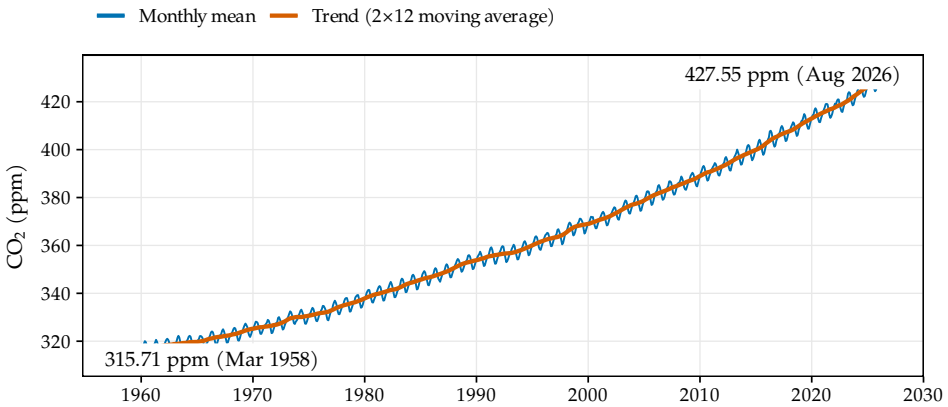
1. Characterize the trend and seasonality.
2. Build and validate an adequate model.
3. Test whether growth is accelerating.
4. Forecast to 2030 with prediction intervals.

The data: Mauna Loa monthly CO_2

- Mauna Loa Observatory, Hawaii: the longest continuous direct CO_2 record, started by C. D. Keeling in 1958.
- **Source:** NOAA Global Monitoring Laboratory [2], https://gml.noaa.gov/ccgg/trends/data.html
- **Variable:** monthly mean mole fraction, in parts per million (ppm).
- Real, public and never analyzed by me before; a large sample for time series modeling.

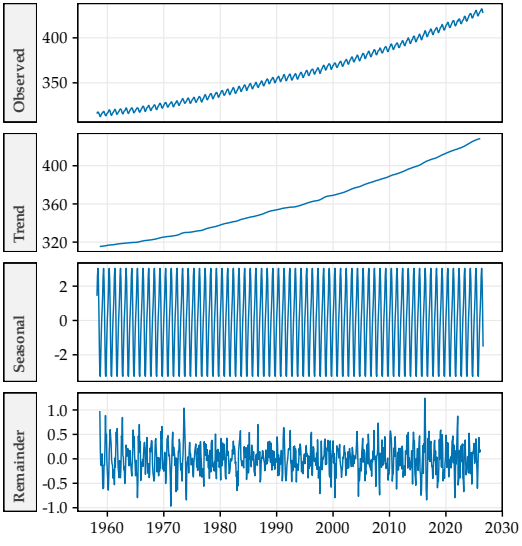
The series at a glance	
Months, n	822
Period	1958–2026
Missing values	none
First value	315.71 ppm
Last value	427.55 ppm
Increase	+111.8 ppm
Relative increase	35%

Step 1: the time plot



Takeaway: An upward trend and a regular 12-month cycle: the mean depends on t, so the series is non-stationary.

Decomposition: trend plus a stable seasonal cycle

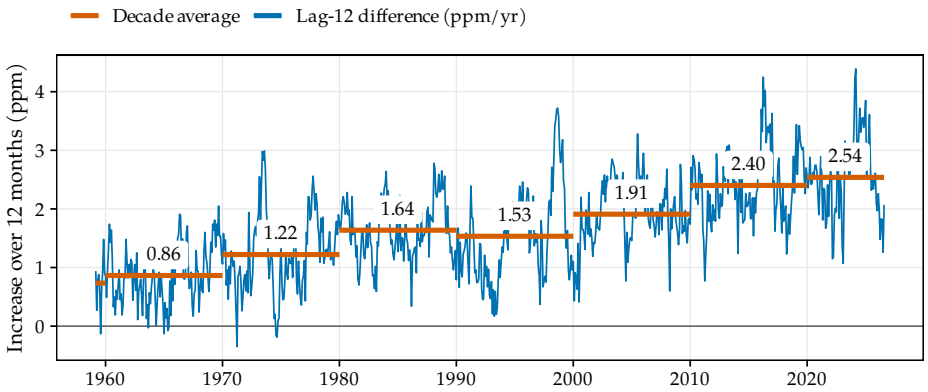


- Additive, 2×12 moving-average filter.
- Trend: 96.6% of the variance.
- Peak in **May** ($+3.04$ ppm), trough in **October** (-3.25 ppm); amplitude **6.29 ppm**.
- Amplitude 6.0 ppm (1960s) vs 7.0 ppm (2010s), while the level rose 35%.

Decision

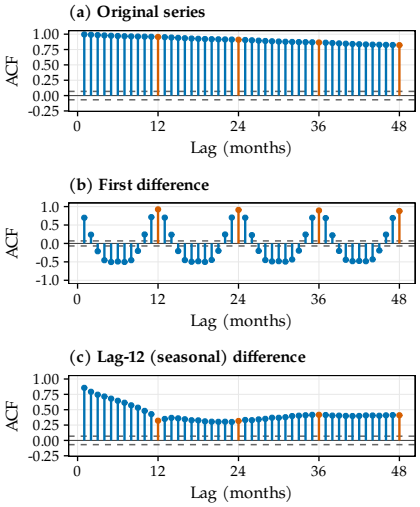
Swings do not grow with the level $\Rightarrow$ **additive model, no transformation**.

The growth rate is accelerating



Takeaway: Growth tripled: 0.86 ppm/yr (1960s) → 2.54 ppm/yr (2020s). Record jumps in El Niño years (2016, 2024).

Achieving stationarity



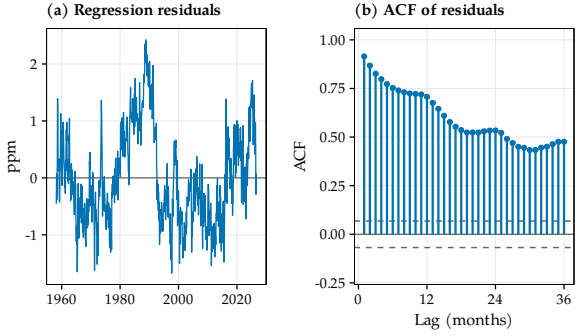
- X_t: ACF decays very slowly ($\hat{\rho}(1) = 0.996$).
- ∇X_t: ACF ≈ 0.9 at lags 12, 24, 36 $\Rightarrow$ seasonal difference.
- $\nabla_{12} X_t$: slow decay $\Rightarrow$ regular difference.

Augmented Dickey–Fuller, H_0: unit root

Series	Type	τ	Decision
Seas. adj. X_t	mean	3.57 to 4.85	keep H_0
Seas. adj. X_t	trend	-0.88 to -0.62	keep H_0
$\nabla\nabla_{12} X_t$	mean	-37.9 to -22.1	reject

$\Rightarrow$ $W_t = (1 - B)(1 - B^{12})X_t$ is **stationary**.

Estimation versus differencing



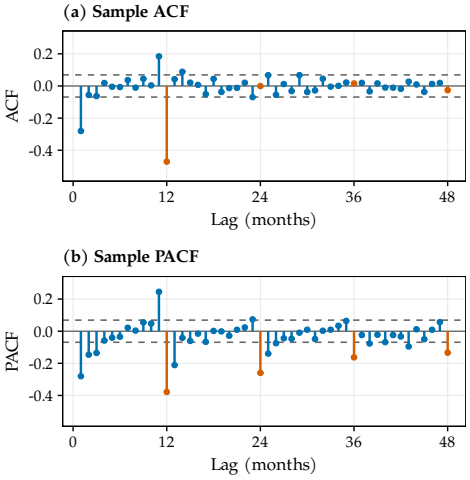
Quadratic trend + 11 monthly dummies:

- Fit: $R^2 = 0.9994$.
- Residual ACF(1) $= 0.915$; $Q(12) = 5991$.
- Durbin–Watson $D = 0.17$ ($\ll 2$).
- ARMA errors: AR root 1.02.

Conclusion

Deterministic trend fails: **difference instead**.

Identification from the ACF and PACF of W_t



- $Q(12) = 283.3$, $p < 0.001$: W_t is not white noise.
- **ACF cuts off** after lags 1 (-0.28) and 12 (-0.47).
- **PACF tails off** at lags 1–3 and 12, 24, 36.
- $\Rightarrow$ MA(1) $\times$ seasonal MA(1).

ARIMA$(0,1,1) \times (0,1,1)_{12}$

$$(1-B)(1-B^{12})X_t = (1-\theta_1 B)(1-\Theta_1 B^{12})Z_t$$

Model selection: AIC and SBC

Model	Parameters	AIC	SBC	ΔSBC	Ljung–Box $Q(24)$ p
ARIMA$(0,1,1)\times(0,1,1)_{12}$	**2**	451.30	**465.39**	**0.00**	0.532
ARIMA$(1,1,1)\times(0,1,1)_{12}$	3	**447.58**	466.37	0.98	0.769
ARIMA$(0,1,2)\times(0,1,1)_{12}$	3	448.59	467.38	1.99	0.723
ARIMA$(0,1,1)\times(1,1,1)_{12}$	3	453.30	472.08	6.69	0.472
ARIMA$(2,1,0)\times(0,1,1)_{12}$	3	460.64	479.42	14.03	0.213
ARIMA$(1,1,0)\times(0,1,1)_{12}$	2	472.83	486.92	21.53	0.038
ARIMA$(0,1,1)\times(1,1,0)_{12}$	2	669.34	683.43	218.04	< 0.001

- Nine candidates fitted by maximum likelihood; **SBC** preferred in this large sample.
- Pure AR and seasonal-AR alternatives fit clearly worse.

Four model families compared

Family	Best model (by SBC)	ACF lag 12	$Q(12)$ p	White noise?
ARMA	ARMA(3,2) for W_t	-0.409	< 0.001	No
ARIMA	ARIMA(4,1,4) with drift	0.180	< 0.001	No
Regression + ARMA errors	ARMA(2,1) errors	0.138	< 0.001	No
Seasonal ARIMA	**ARIMA**$(0,1,1) \times (0,1,1)_{12}$	$\mathbf{-0.005}$	**0.464**	**Yes**

- Non-seasonal ARIMA mimics the seasonal difference with AR roots *on* the unit circle (periods 6 and 12).
- Only the seasonal ARIMA captures the seasonal structure, with just **two parameters**.

Estimates, hypothesis tests and confidence intervals

Method	Parameter	Estimate	SE	z	p	95% CI
Maximum likelihood	θ_1	0.3885	0.0367	10.59	< 0.001	[0.317, 0.460]
	Θ_1	0.8700	0.0170	51.28	< 0.001	[0.837, 0.903]
Conditional least squares	θ_1	0.4172	0.0360	11.60	< 0.001	[0.347, 0.488]
	Θ_1	0.8116	0.0194	41.84	< 0.001	[0.774, 0.850]

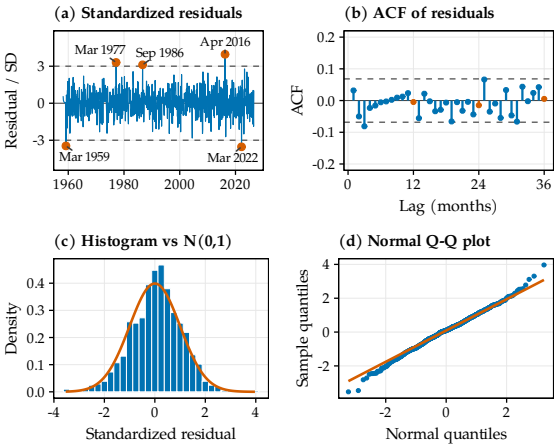
Inference

Both parameters highly significant; ML and CLS agree. $\hat{\sigma}_Z = 0.318$ ppm.
Invertible: roots 2.57 and $1.012 > 1$.

Interpretation

$\hat{\theta}_1$: 39% of a monthly shock reverses next month.
$\hat{\Theta}_1 = 0.87$: very stable seasonal pattern; CI excludes 1.

Model adequacy: residual diagnostics



Test	Statistic	p
Ljung–Box $Q(6)$	9.05	0.060
Ljung–Box $Q(12)$	9.74	0.464
Ljung–Box $Q(18)$	14.43	0.567
Ljung–Box $Q(24)$	20.83	0.532
$Q(12)$, squared resid.	12.40	0.414
Shapiro–Wilk	0.996	0.048

- Independence ✓ Constant variance ✓
- Normality marginal: 5 months beyond ± 3.

Remedial measures

1. Small lag-3 autocorrelation

Added an AR(1) term: $\hat{\phi}_1 = 0.215$ ($p = 0.010$).

- SBC rises (466.37 vs 465.39).
- Hold-out RMSE worse: 1.168 vs 1.110.
- Interval coverage only 83% vs 97%.

$\Rightarrow$ **Rejected**: keep the parsimonious model.

2. Five outlying months

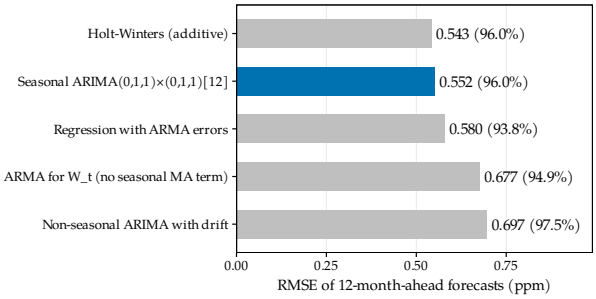
Re-estimated with those months set to missing:

- $\hat{\theta}_1 = 0.384$, $\hat{\Theta}_1 = 0.869$: **unchanged**.
- Normality restored: Shapiro–Wilk $p = 0.211$, Anderson–Darling $p = 0.101$.

$\Rightarrow$ Inference is **robust**.

Takeaway: ARIMA$(0,1,1) \times (0,1,1)_{12}$ is adequate; no transformation is needed.

Forecast evaluation

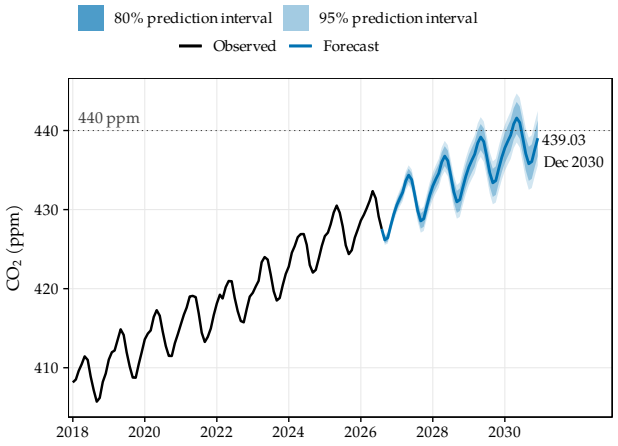


Rolling origin: refit each August 2000–2022; forecast 12 months (276 forecasts).

- Seasonal ARIMA: RMSE 0.552 ppm, 95% coverage 96.0%.
- Best of the four model families; tied with Holt–Winters smoothing.

Hold-out (Sep 2023 – Aug 2026): RMSE 1.110 ppm, coverage 97.2%.

Forecasts to 2030



April 2030

CO_2 first exceeds **440 ppm**

440.84 ppm [437.78, 443.89]

- Dec 2030: 439.03 ppm
  95% PI [435.59, 442.47]
- Growth $\approx$ **2.4 ppm/yr**
- Holt–Winters agrees within 0.13 ppm.

Conclusions

1. CO_2 is non-stationary with an **accelerating trend** and a **stable 12-month cycle**; regular and seasonal differencing achieve stationarity.
2. **ARIMA**$(0,1,1) \times (0,1,1)_{12}$ is selected by ACF/PACF identification and SBC; both parameters are highly significant and the model is invertible.
3. Residuals are white noise with constant variance; inference is robust to five outlying months.
4. Best of four model families in out-of-sample forecasting (RMSE 0.55 ppm, 96% coverage).
5. CO_2 rises ≈ 2.4 ppm/yr and is forecast to pass 440 ppm in April 2030.

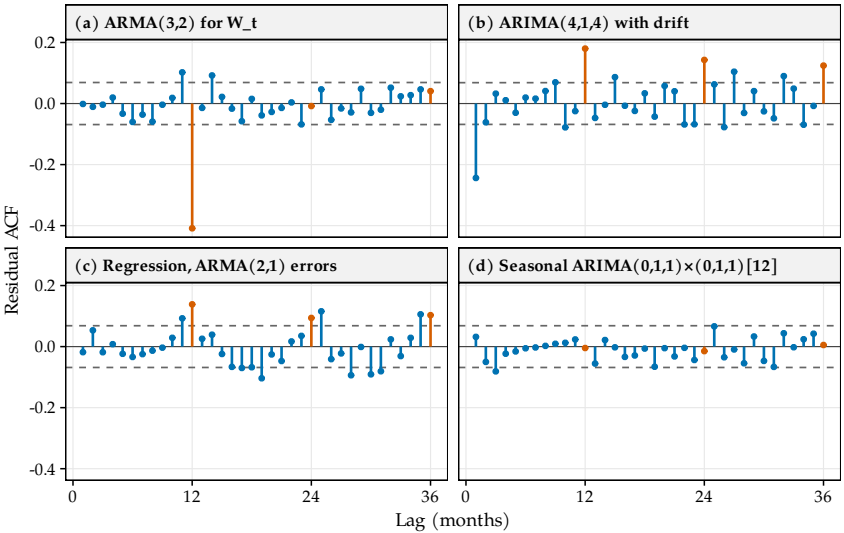
Limitations and extensions

Univariate model: it cannot anticipate El Niño surges (2016, 2024). Adding climate indices or emissions data is a natural extension.

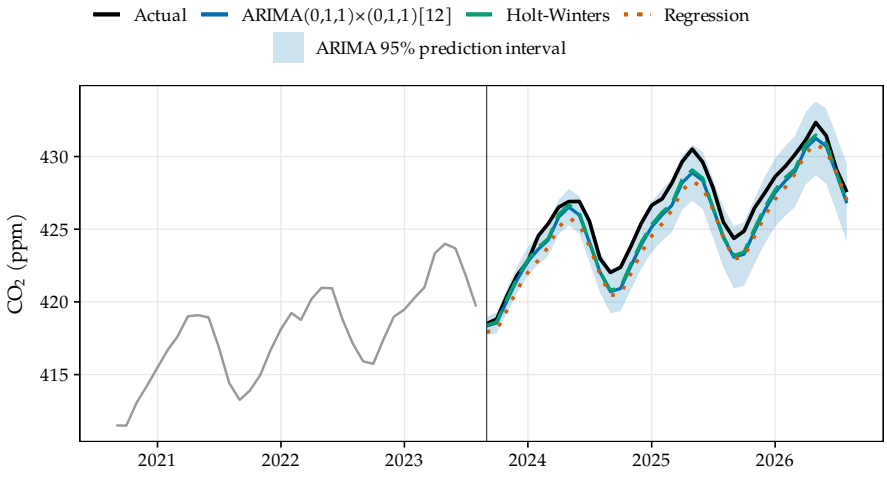
Thank you

Questions?

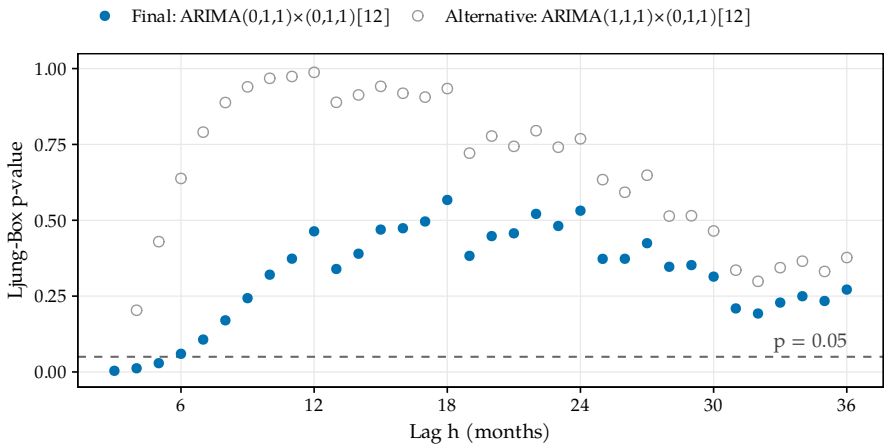
Backup: residual ACF of the four model families



Backup: 36-month hold-out forecasts



Backup: Ljung–Box p-values by lag



References

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